

AQI Forecast Dashboard

Real-time Air Quality Index Prediction

User Manual

End-User Guide for the Streamlit Dashboard

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Powered by: LightGBM + OpenAQ v3 + CPCB Standards
Frontend: Streamlit
Backend: FastAPI · <http://localhost:8000>

Supported Cities: Chennai · Delhi · Mumbai · Kolkata · Ahmedabad

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1. Getting Started

1.1. What is this Dashboard?

AQI Forecast Dashboard — Overview

This dashboard predicts the **Air Quality Index (AQI)** for Indian cities using live sensor data from CPCB-registered monitoring stations via the OpenAQ v3 API, combined with a LightGBM machine learning model trained on 25 Indian cities.

You get:

- Current AQI with health category
- 24-hour AQI forecast
- Live pollutant levels — PM2.5, PM10, NO₂, SO₂, CO, O₃
- 7-day historical trend
- Plain-language health advisory



Figure 1: AQI Forecast Dashboard — full view after a prediction

1.2. Step-by-Step: Get Your First Prediction

Follow these five steps to get your first AQI forecast.

1. Open the dashboard

Navigate to the dashboard URL in your browser:

<http://localhost:8501>

2. Filter by city

Use the **Filter by city** dropdown in the left sidebar to narrow the station list to your desired city.

3. Select a station

Choose a monitoring station from the **Select station** dropdown. Format: **City · Station Name (ID)**

4. Click Predict AQI

Click the **Predict AQI** button. The system fetches live data from OpenAQ — allow **10–60 seconds** on first run.

5. Read your results

The dashboard displays: current AQI, 24-hour forecast chart, pollutant levels, 7-day historical trend, and a health advisory.

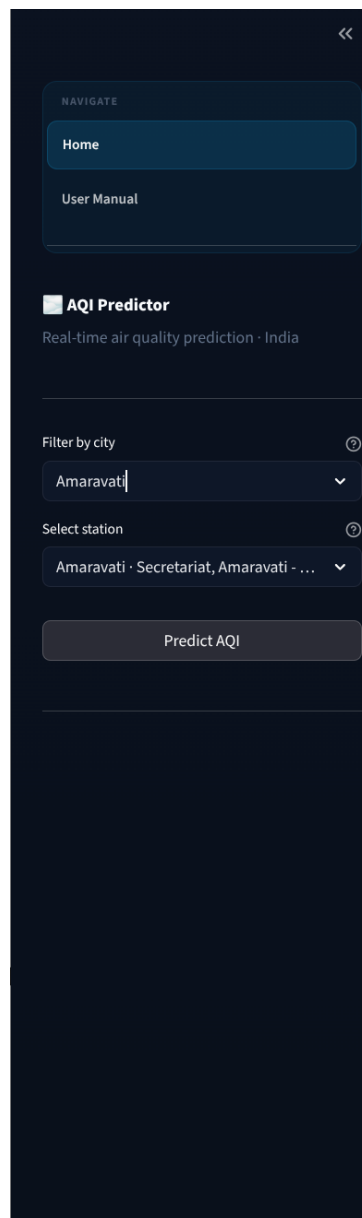


Figure 2: Sidebar: city filter and station selector

2. Dashboard Guide

2.1. Understanding Each Section

2.1.1. AQI Hero Card

AQI Hero Card

- Large coloured number = **current AQI**
- Colour shifts green → yellow → red with severity
- Click [What does this AQI mean?](#) for the full scale popup
- **Primary Pollutant** = the gas most responsible for the current AQI
- **Last Updated** shows sensor reading time in IST

2.1.2. 24-Hour Forecast Chart

24-Hour Forecast Chart

- **X-axis:** H+1 to H+24 = next 24 hours
- **Y-axis:** predicted AQI value
- Dotted horizontal line = current AQI (your baseline reference)
- Forecast above the line ⇒ air quality is worsening
- Each dot is coloured by AQI severity category

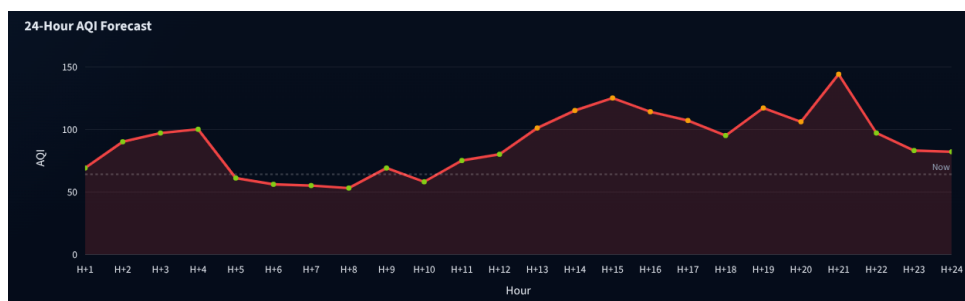


Figure 3: 24-hour forecast — each dot coloured by AQI severity

2.1.3. 7-Day Trend

7-Day Historical Trend

- Shows daily average AQI over the past 7 days
- Each point is coloured by its AQI category
- A rising trend = worsening air quality this week
- Dates are shown in IST format

2.1.4. Pollutant Levels Chart

Pollutant Levels Bar Chart

- Horizontal bar chart of all measured gases
- Each pollutant has a distinct colour
- Longer bar = higher concentration
- Units: $\mu\text{g}/\text{m}^3$ for PM, NO_2 , SO_2 , O_3 · mg/m^3 for CO

2.2. Forecast Summary Card

Field	Description
Current AQI	Value from the latest sensor reading
24h Peak	Highest predicted AQI in the next 24 hours
24h Average	Mean predicted AQI across the 24-hour forecast window
Primary Pollutant	The dominant gas measured at this station right now

Table 1: Forecast Summary Card fields

3. AQI Scale

3.1. CPCB National AQI Colour Scale

India uses the **CPCB (Central Pollution Control Board)** AQI scale. Each category has a colour, a numeric range, and a health guideline.

Range	Category	Colour	Health Impact	Recommended Action
0–50	Good	Green	Minimal health impact.	Safe for all outdoor activities.
51–100	Satisfactory	Lime	Minor discomfort to sensitive people.	Generally safe. Sensitive groups take care.
101–200	Moderate	Amber	Discomfort for lung and heart patients.	Reduce prolonged outdoor activity.
201–300	Poor	Orange	Discomfort for most people.	Avoid prolonged outdoor exposure.
301–400	Very Poor	Red	Respiratory illness risk.	Stay indoors if possible.
401–500	Severe	Dark Red	Serious risk for everyone.	Avoid all outdoor activity.

Table 2: CPCB National Air Quality Index Scale — source: airquality.cpcb.gov.in



Figure 4: PCB AQI reference strip as shown at the bottom of the dashboard

3.2. How to Read the Popup on the Dashboard

On the main dashboard, below the AQI hero number, click the button:

What does this AQI value mean?

A popup opens showing the full CPCB scale with your **current category highlighted** with a “● **Current**” badge, so you can immediately see where your station’s air quality stands.

4. Pollutant Reference Guide

These are the six pollutants measured by CPCB stations and displayed on the dashboard.

Symbol	Full Name	Unit	Primary Source	Health Concern
PM2.5	Fine Particulate Matter	$\mu\text{g}/\text{m}^3$	Vehicles, industry, biomass burning	Penetrates deep into lungs and bloodstream. Most harmful pollutant for health.
PM10	Coarse Particulate Matter	$\mu\text{g}/\text{m}^3$	Road dust, construction, wind	Affects nose, throat, and upper airways. Less penetrating than PM2.5.
NO₂	Nitrogen Dioxide	$\mu\text{g}/\text{m}^3$	Vehicle exhaust, power plants	Irritates airways, worsens asthma and respiratory disease.
SO₂	Sulphur Dioxide	$\mu\text{g}/\text{m}^3$	Coal burning, industrial processes	Causes breathing difficulty and eye irritation. Reacts to form acid rain.
CO	Carbon Monoxide	mg/m^3	Incomplete combustion, vehicles	Reduces oxygen delivery in blood. High levels can be fatal.
O₃	Ground-level Ozone	$\mu\text{g}/\text{m}^3$	Sunlight reacting with NOx and VOCs	Irritates airways, damages lung tissue. Worse on hot sunny days.

5. Troubleshooting

5.1. Common Issues & Fixes

Not enough recent data for this station

Cause: This station hasn't reported 7 days of readings yet. The forecast model needs at least 168 hours of sensor history.

Fix: Try a different station in the same city, or check back the next day.

Service is currently offline

Cause: The backend prediction service is not running or unreachable.

Fix: If you are an admin: run `docker compose up -d` and refresh the page.

Prediction is taking very long (> 60 seconds)

Cause: The system fetches fresh live data from OpenAQ on every prediction. First predictions for a station can take up to 60 seconds.

Fix: Click **Predict AQI** again if it times out. Subsequent runs are faster.

This city is not available yet

Cause: This city was not included in the model's training dataset.

Fix: Please select a different station from the available list.

Charts are not showing

Cause: Forecast or trend data was not available for this station.

Fix: Try a different station that has more historical sensor readings.

5.2. Data Sources

Source		Description
OpenAQ API	v3	Live hourly pollutant readings from CPCB-registered monitoring stations across India
Open-Meteo API		Weather features — temperature, humidity, wind speed, and atmospheric pressure
CPCB Standards	Stan-	AQI calculation methodology and health category thresholds
LightGBM Model		Gradient boosting ML model trained on 25 Indian cities · 24-hour forecast horizon
Apache Airflow		Hourly automated data pipeline that fetches, cleans, and stores sensor readings

Table 4: Data sources powering the AQI Forecast Dashboard